

REQUIREMENT DOCUMENT

1.1 Purpose

The purpose of this Employee Management System (EMS) is to provide a platform for storing and managing employee information.

The system provides:

- Role-based access control to protect information
- Provide a secure login and authentication mechanism
- Ensure event logging
- Prevent unauthorized or invalid operations through a set of predefined error codes

1.2 Users and Roles

1. Root

- Highest-level system administrator
- Can view, update, and delete any employee record

2. Management Staff

- Manages employee data
- Permissions include:
 - Adding new employees
 - Removing existing employees
 - Updating employee position and salary
 - Viewing all employee details including salary
 - Searching for employees by ID or position

3. Hardware Engineer

- Has limited access
- Permissions include:
 - Viewing assigned hardware units
 - Updating hardware unit details
 - Searching and viewing employee data
 - Cannot view salary information
 - Cannot add, remove, or update employees

4. Software Engineer

- Has limited access
- Permissions include:
 - Searching and viewing employee data
 - Cannot view salary information
 - Cannot add, remove, or update employees

1.3 Employee management features

- View the employee details (By category, Position) and total number of employees.
- Add a new employee.
- Remove an existing employee by ID.
- Update an employee's position.
- Update an employee's salary.
- Search employee details by ID or by Position.

1.4 Event logging

Provides multiple log levels like

- INFO - Normal, successful operations
- WARNING - Invalid actions that are not critical
- ERROR - Unauthorized access attempts

1.5 Error codes

- 1. 9000 – Creation of a duplicate instance of the Employee Management System.
- 2. 9001 – Attempt to check or update hardware count for a non-hardware engineer.
- 3. 9002 – Employee ID not found during search, update, or removal.
- 4. 9003 – Unauthorized access attempt (failed login).